Trainings

Preparatory Steps

Industrial and Power Generation

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first, and attach the negative (-) battery cable last.

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, or prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

- Disconnect the batteries to prevent accidental starting. See equipment manufacturer service information.
- Disconnect the air starter, if equipped, to prevent accidental starting. See equipment manufacturer service information.
- Remove the starting motor. Refer to Procedure 013-020 in Section 13. (/qs3/pubsys2/xml/en/procedures/56/56-013-020-tr.html)
- Remove the engine speed sensor. Use the following procedure in the QSK19, QSK23, QSK45, QSK60, and QSK78 Electronic Control System Troubleshooting and Repair Manual, Bulletin 3666113. Refer to Procedure 019-042 in Section 19. (/qs3/pubsys2/xml/en/procedures/19/19-019-042.html)
- Remove the transmission, clutch, and all related components. See equipment manufacturer service information.

- Remove the flywheel. Refer to Procedure 016-005 in Section 16. (/qs3/pubsys2/xml/en/procedures/56/56-016-005-tr.html)
- Drain the engine lubricating oil. Refer to Procedure 007-037 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-037-tr.html)
- Remove the rear crankshaft seal. Refer to Procedure 001-024 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-024-tr.html)

with Electronically Actuated Injector

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- Remove the engine speed sensor. Use the following procedure in the QSK38, QSK50 and QSK60 (CM850 Modular Common Rail System) Troubleshooting and Repair Manual, Bulletin 4021533. Refer to Procedure 019-042 in Section 19. (/qs3/pubsys2/xml/en/procedures/05/05-019-042.html)
- Remove the rear bank wiring harness. Use the following procedure in the QSK38, QSK50 and QSK60 (CM850 Modular Common Rail System) Troubleshooting and Repair Manual, Bulletin 4021533. Refer to Procedure 019-043 in Section 19. (/qs3/pubsys2/xml/en/procedures/05/05-019-043.html)
- Remove the turbocharger oil drain lines. Refer to Procedure 010-034 in Section 10. (/qs3/pubsys2/xml/en/procedures/56/56-010-034-tr.html) Refer to Procedure 010-035 in Section 10. (/qs3/pubsys2/xml/en/procedures/56/56-010-035-tr.html)

- Remove the transmission, clutch, and all related components. See equipment manufacturer service information.
- Remove the flywheel. Refer to Procedure 016-005 in Section 16. (/qs3/pubsys2/xml/en/procedures/56/56-016-005-tr.html)
- Drain the engine lubricating oil. Refer to Procedure 007-037 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-037-tr.html)
- Remove the rear crankshaft seal. Refer to Procedure 001-024 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-024-tr.html)

QSK60 Marine Applications

⚠ WARNING ⚠

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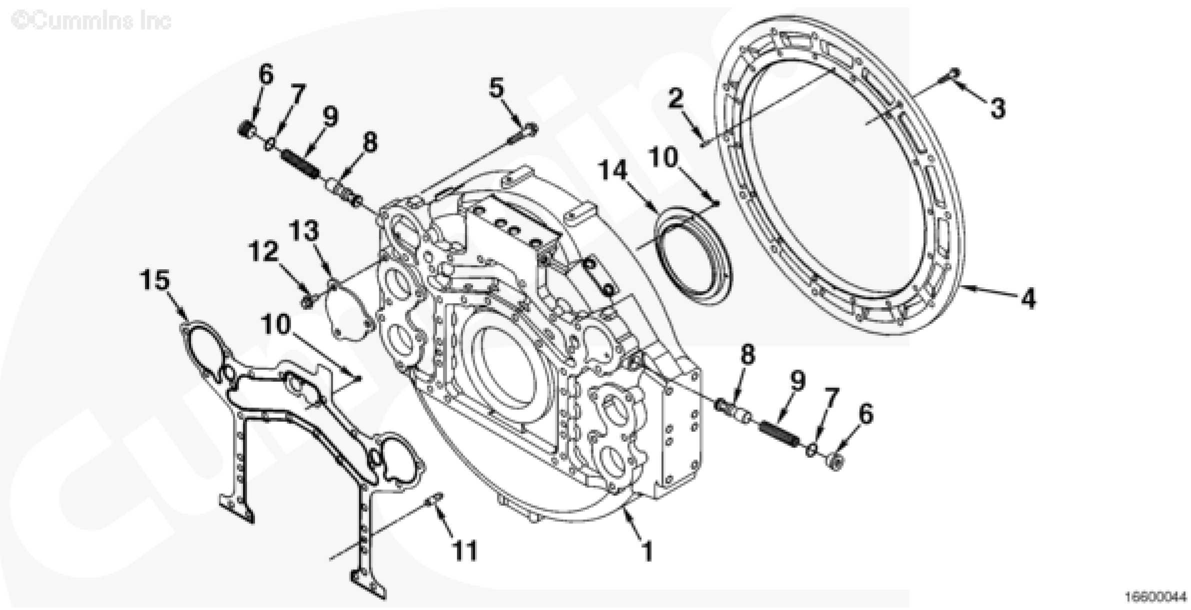
- Disconnect the batteries to prevent accidental starting. See equipment manufacturer service information.
- Disconnect the air starter, if equipped, to prevent accidental starting. See equipment manufacturer service information.
- Remove the starting motor. Refer to Procedure 013-020 in Section 13. (/qs3/pubsys2/xml/en/procedures/56/56-013-020-tr.html)
- Remove the turbocharger air inlet connections. Refer to Procedure 010-130 in Section 10. (/qs3/pubsys2/xml/en/procedures/56/56-010-130-tr.html)
- Remove the air cleaner assembly. Refer to Procedure 010-013 in Section 10. (/qs3/pubsys2/xml/en/procedures/56/56-010-013-tr.html)

- Remove the transmission, clutch, and all related components. See equipment manufacturer service information.
- Remove the flywheel. Refer to Procedure 016-005 in Section 16. (/qs3/pubsys2/xml/en/procedures/56/56-016-005-tr.html)
- Drain the engine lubricating oil. Refer to Procedure 007-037 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-037-tr.html)
- Remove the rear crankshaft seal. Refer to Procedure 001-024 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-024-tr.html)

Exploded View



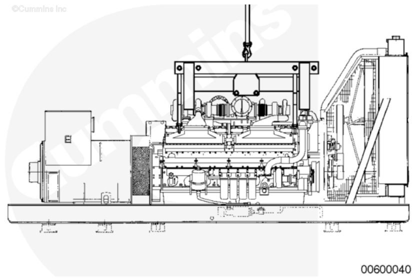
Flywheel Housing

1. Flywheel housing
2. Transmission adapter locating roll pin
3. Capscrew (M12 x 1.75 x 50)
4. Transmission adapter
5. Capscrew
6. Pressure regulator plug
7. O-ring seal
8. Pressure regulator plunger for piston cooling nozzles
9. Pressure regulator spring
10. Capscrew (M6 x 1.00 x 12)
11. Flywheel housing locating dowel pin
12. Capscrew (M6 x 1.0 x 20)
13. Access hole cover
14. Rear crankshaft oil seal
15. Flywheel housing mounting gasket.

Remove

⚠ WARNING ⚠

Inspect the engine lifting brackets for damage or cracks. Do not attempt to lift the engine if any cracks or damage is visible. Using damaged lifting equipment can cause serious personal injury and or property damage.



⚠ WARNING ⚠

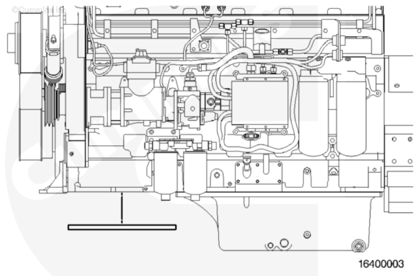
The QSK45 and QSK60 series engines can weigh as much as 9980 kg [22,000 lb]. Use a properly rated hoist and lifting fixture, Part Number 3163264, to lift the engine. Rigging and lifting must be done by trained, experienced personnel.

Install the engine lifting fixture.

Remove the capscrews from the engine mounts.

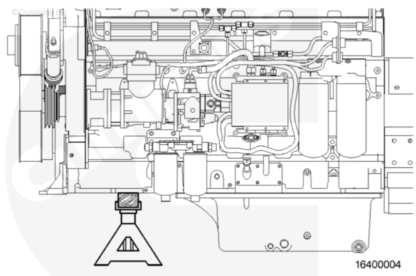
Use the engine lifting fixture to lift the engine up until the engine mounts are free from the frame.

Remove the adapter cover plate or oil sump, whichever is in the front position.



⚠ CAUTION ⚠

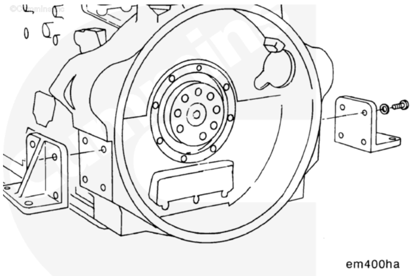
Put a wooden block the width of the oil pan adapter between the jack stands and the oil pan adapter to prevent damage to the engine. Use a minimum of four jack stands to support the engine.



Each jack stand **must** be rated at a minimum of 2,722 kg [6,000 lbs].

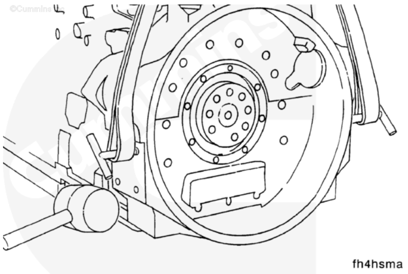
Use the engine lifting fixture to lift the engine and support the front and rear on both sides of the engine with jack stands and wooden blocks.

Remove the rear engine mounts from the flywheel housing.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Remove two capscrews and install the guide studs.

Use a hoist, T-handles, and a lifting sling to remove the flywheel housing.

Install the lifting sling, and adjust the hoist until there is tension in the lifting sling.

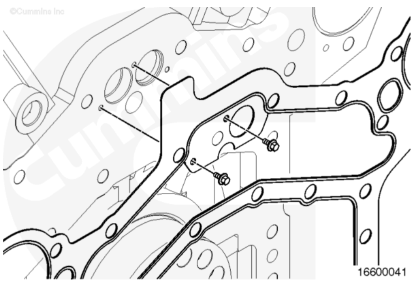
Remove the remaining flywheel housing mounting capscrews.

Use a mallet to tap the flywheel housing off the two locating dowel pins.

Note the gasket orientation. Engine damage can occur if the gasket is **not** properly installed when installing the flywheel housing.

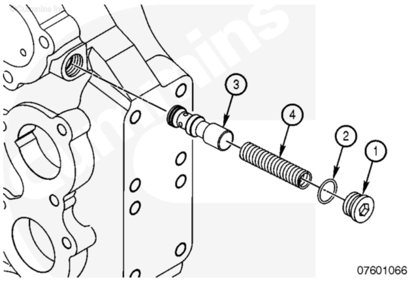
Remove the gasket.

Remove the two capscrews securing the gasket in place to the engine block. Remove the gasket.



⚠ WARNING ⚠

Wear appropriate eye and face protection when performing this procedure. Springs are under tension and can act as a projectile if released, causing personal injury.



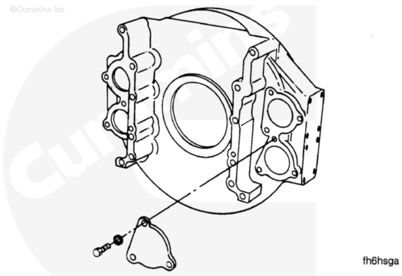
Remove the pressure regulator from the flywheel housing.

Remove the plug (1), springs (4), and plunger (3) from the flywheel housing. Remove and discard the o-ring (2) from the plug.

Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use solvent or steam to clean the flywheel housing.

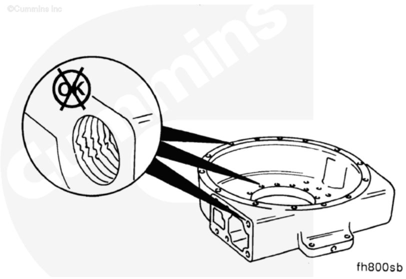
Dry with compressed air.

Inspect all surfaces for nicks, burrs, cracks, or other damage.

Use a fine crocus cloth to remove small nicks and burrs.

Inspect all threaded capscrew holes for damage.

Repair or replace the housing if the capscrew holes are damaged.



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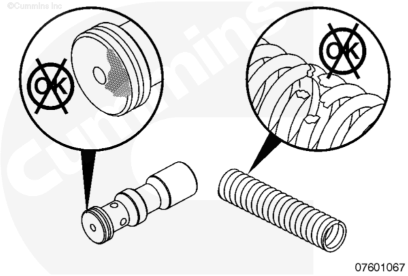


⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

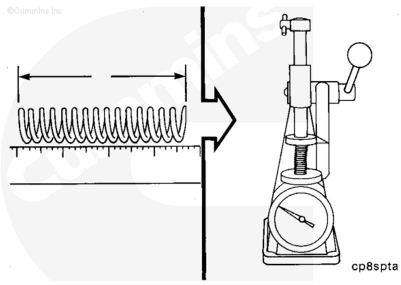
Use solvent to clean the lubricating oil pressure regulator spool and spring. Dry with compressed air.

Inspect the spring for broken coils.

Inspect the spool for scuffing. Minor scuffs can be removed with a fine crocus cloth.

The critical area of the spool is the seating area. Polished areas are normal.

Use a Cummins® spring tester, Part Number 3375182, to measure the pressure regulator spring force.



Compress the spring to 114.3 mm [4.50 in].

The force required to compress the pressure regulator spring **must** be:

Force Required to Compress Pressure Regulator Spring		
N	**Assembled Length**	**ft-lb**
112.5	114.3 mm [4.50 in]	25.3

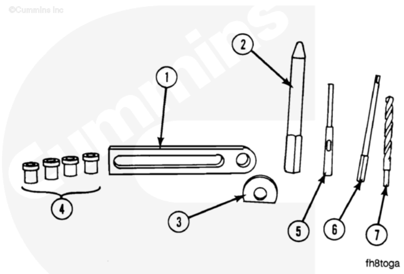
If the pressure regulator spring does **not** meet this specification, it **must** be replaced.

Redowel

The tools needed to perform this procedure are:

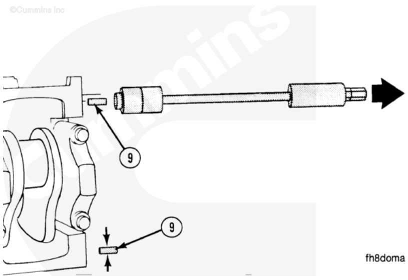
Cummins® drill ream fixture, Part Number ST-1232, that contains:

1. Cummins® plate, Part Number ST-1232-1
2. Cummins® locator pin, Part Number 3375052
3. Cummins® spacer washer, Part Number ST-1232-2
4. Drill/ream actual sizes bushing set depend on the dowel size
5. Drill adapter locally obtained; use to adapt open-shank reamers to drill-chuck
6. Reamer locally obtained
7. Drill bit locally obtained.



Use a Cummins® dowel pin extractor, Part Number ST-1134, or equivalent. Remove the two dowels (9) from the block.

Measure a dowel pin that has been removed so an oversize dowel pin dimension can be determined.



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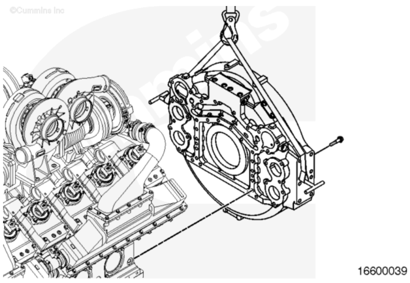


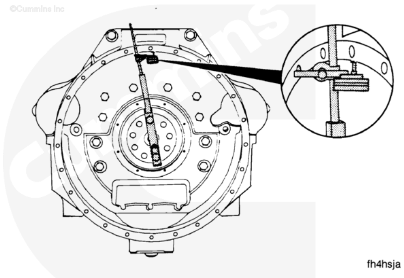
Install the edge-molded gasket in the correct position.

Use guide bolts to help during alignment. Install the flywheel housing and the capscrews.

Install the flywheel housing.

Do **not** tighten the capscrews. The flywheel housing will have to be aligned.

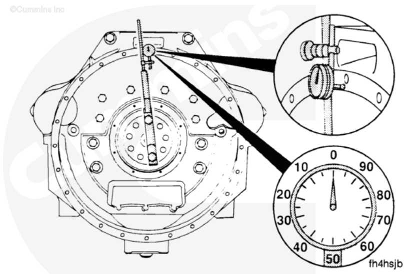
Move the housing with a mallet until the bore is within specifications. Make sure that the face of the housing is in alignment.



Flywheel Housing Bore Diameter			
	mm		in
Number 0	648	NOM	25.5
Number 00	787	NOM	31.0

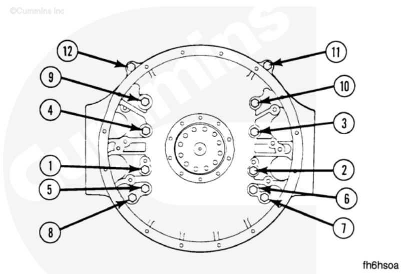
Flywheel Housing - Radial Runout				
	mm		in	
Number 0		0.25	MAX	0.010
Number 00		0.30	MAX	0.012

Measure the flywheel housing face runout.



Flywheel Housing - Face Runout				
	mm		in	
Number 0		0.25	MAX	0.010
Number 00		0.30	MAX	0.012

When the housing is in alignment, tighten the capscrews in the sequence shown.

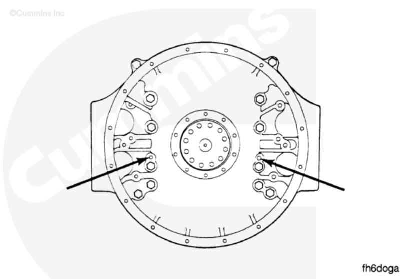


Torque Value:

1. 95 n•m [70 ft-lb]
2. 195 n•m [144 ft-lb]

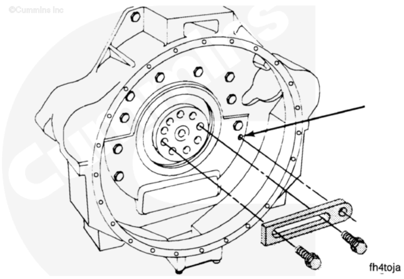
After the capscrews are tightened, check the alignment again.

The graphic illustrates the location of the dowel holes on QSK45 and QSK60 flywheel housings.



Use the appropriate capscrews to attach the Cummins® plate, Part Number ST-1232-1, that is contained in the Cummins® drill/ream fixture, Part Number ST-1232, to the crankshaft.

Do **not** completely tighten the capscrews. The plate **must** be able to move.



Use the locator pin to align the plate with the hole for the dowel pin.

The taper on the pin **must** engage the dowel pin hole.

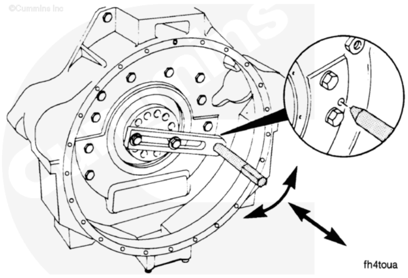
Tighten the capscrews.

The locator pin **must** rotate easily after the capscrews are tightened.

Lock the crankshaft into position.

Make sure the locator pin is still in alignment and that the locator pin can be rotated easily.

The crankshaft **must not** turn during the reaming process.

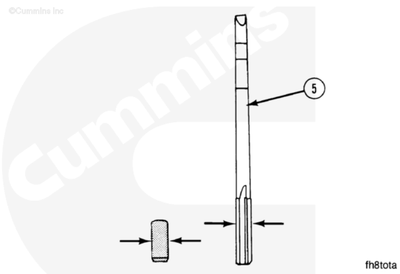


Measure the dowel pins to be installed.

Obtain a reamer (5) that is 0.013 to 0.025 mm [0.0005 to 0.001 in] smaller than the dowel.

The dowel **must** protrude from the block by one-half of the flywheel housing wall thickness.

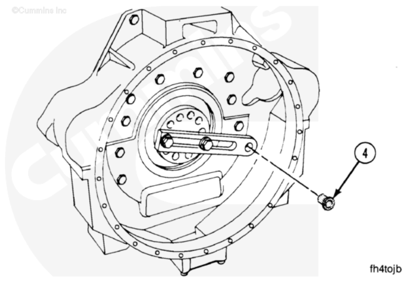
There are three oversize dowel pins available from Cummins Inc.



Measurements			
		mm	in
Oversize Dowel Pin - Outer Diameter	Oversize 0.381 mm [0.015 in]	13.08	0.515
	Oversize 0.762 mm [0.030 in]	13.46	0.530

Measurements			
		mm	in
	Oversize 1.143 mm [0.045 in]	13.84	0.545

Install the appropriate drill bushings (4). The following bushings are available from Cummins Inc.



Available Drill/Ream Bushing Sets	
Part Number	**Bushing Sizes**
3376495	1.230 mm [0.484 in], 1.270 mm [0.500 in], 13.106 mm [0.516 in], 13.487 mm [0.531 in], 13.868 mm [0.546 in]
ST-1234	14.300 mm [0.563 in], 14.681 mm [0.578 in], 15.088 mm [0.594 in], 15.487 mm [0.694 in],
ST-1235	15.875 mm [0.625 in], 16.281 mm [0.640 in], 16.662 mm [0.656 in], 17.069 mm [0.672 in]
ST-1236	17.475 mm [0.688 in], 17.856 mm [0.703 in], 18.263 mm [0.719 in], 18.644 mm [0.734 in]
ST-1237	19.050 mm [0.750 in], 19.456 mm [0.766 in], 19.837 mm [0.781 in]
ST-1238	22.631 mm [0.891 in], 23.825 mm [0.938 in]

The drill bushing used **must** be the same size as the reamer (or the drill) used.

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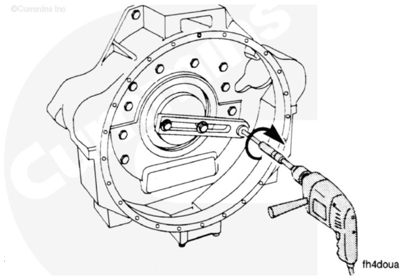


⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

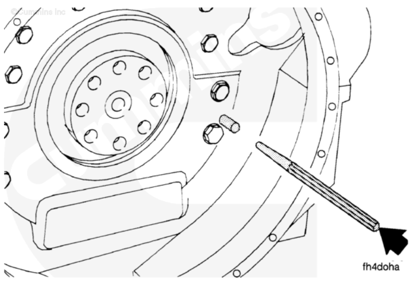
Do not allow metal chips to enter the engine. Damage to the engine will result.

When the new dowel pins are more than 0.38 mm [0.015 in] larger than the old dowels, drill the hole to a size that is slightly smaller than the reamer. The reamer will **not** have to remove an excessive amount of material.

Ream the hole until the reamer touches the bottom of the hole in the block.

Remove the reamer, clean the hole with solvent, and dry with compressed air. Push the reamer through the hole again. The reamer **must** touch the bottom of the hole in the block.

After reaming one hole, turn the plate and align it with the next dowel hole. Repeat the procedure in the next hole.

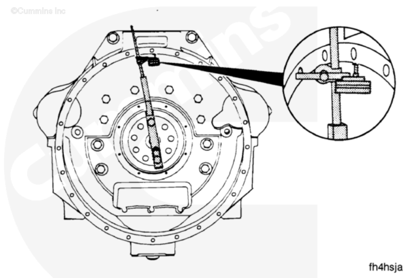


⚠ CAUTION ⚠

The dowel hole must not contain any metal chips. Engine damage will result.

Remove the plate from the crankshaft.

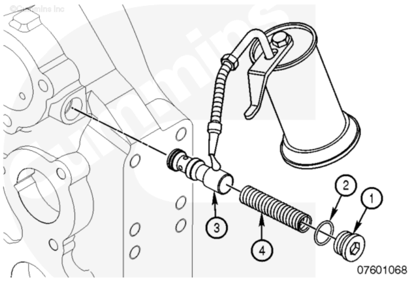
Use a square-nose drift. Drive in each dowel pin until it touches the bottom of the hole in the block.



After the dowels are installed, measure the bore and the face alignment.

If **not** within specifications, repeat the alignment and doweling procedure.

Install



Use clean engine oil to lubricate the oil pressure regulator bores on the flywheel housing.

Install the pressure regulator plunger (3) into the bore with the flat face of the plunger first.

Check that the plunger has freedom of movement the entire length of the bore.

Insert the regulator valve springs (4) into the bore.

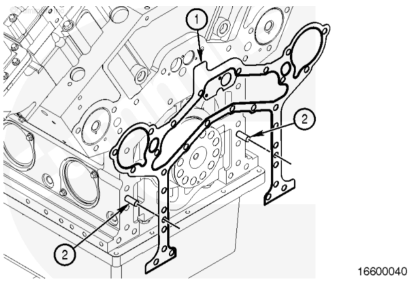
Assemble a new o-ring (2) on the plug (1).

Lubricate the o-ring (2) and compress the pressure regulator spring sufficiently to engage the plug to the threads in the bore. Tighten the plug.

Torque Value:

Oil Pressure Regulator Plug

1. 60 n•m [44 ft-lb]



⚠ CAUTION ⚠

The tab at the top of the flywheel housing gasket is to provide indication of proper orientation of the gasket. Failure to do so will cause excessive lubrication oil temperatures which will lead to engine damage.

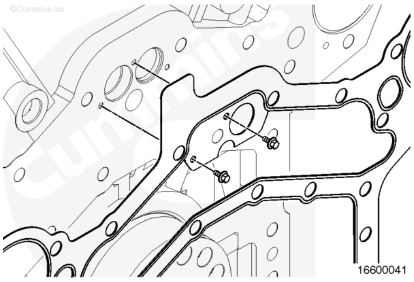
Install the flywheel housing gasket over the dowels (2) on the engine block. Make sure that the tab (1) on the gasket is located on the same side as the lubricating oil filter head assembly. The tab shows which of the two oil cooler rifles will be covered to make sure oil is passed through the oil coolers. Example: If the oil filter head is on the right bank of the engine, the right bank oil cooler rifle needs to be covered. Therefore, the tab (1) on the flywheel housing gasket must be on the right bank side.

Secure the flywheel gasket in place with the two capscrews located below the tab.

Torque Value:

Flywheel Gasket Capscrews

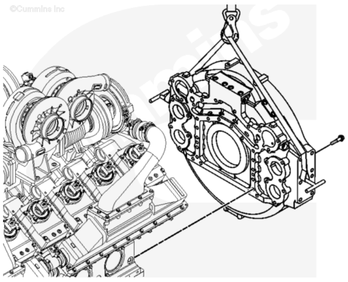
1. 9 n•m [80 in-lb]



16600041

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



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Make sure the flywheel housing dowel pins have been installed into the cylinder block.

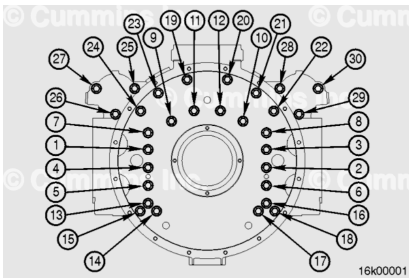
Use guide bolts to help during alignment.

Install the flywheel housing and the capscrews.

Tighten the flywheel housing mounting capscrews in sequence.

Torque Value:

1. 95 n•m [70 ft-lb]
2. 195 n•m [144 ft-lb]



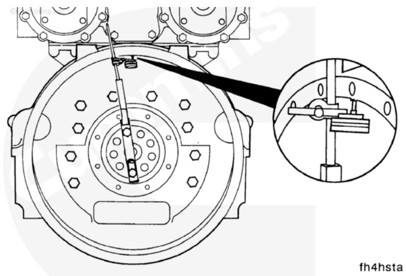
16k00001

Note : There are two types of flywheel housings used on QSK45 and QSK60 engine applications; small (0) and large (00). A small (0) flywheel housing is shown in the illustration. The torque sequence and torque values apply to both flywheel housings.

The bore and the face of the housing **must** align with the crankshaft.

The indicator arm **must** be rigid for an accurate reading. It **must not** sag.

Attach an indicator to the crankshaft as shown.

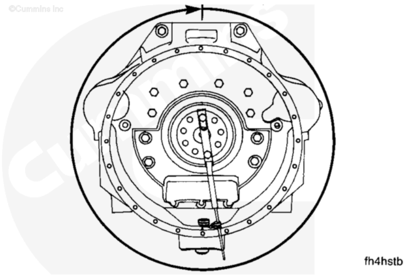


fh4hsta

Position the indicator at the 12-o'clock position. Adjust the dial until the needle points to zero. Rotate the crankshaft one complete revolution, 360 degrees.

Record the total indicator runout.

The indicator **must** return to zero when returning to the start point.



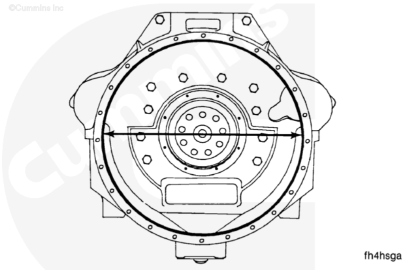
The maximum allowable total indicator runout depends on the diameter of the bore.

Flywheel Housing Bore Diameter			
	mm		in
Number 0	648	NOM	25.5
Number 00	787	NOM	31.0

Flywheel Housing - Radial Runout				
	mm		in	
Number 0		0.25	MAX	0.010
Number 00		0.30	MAX	0.012

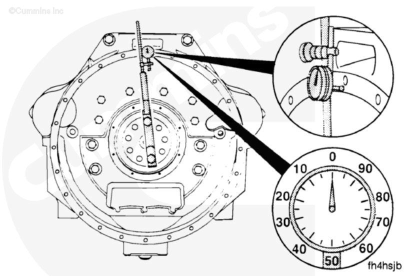
If the alignment is **not** within specifications and the bore is round, the housing can be shifted.

If the alignment is **not** within specifications and the bore is **not** round, the housing **must** be replaced.



When measuring the face runout, the crankshaft **must** be pushed or pulled in the same direction each time a point is measured.

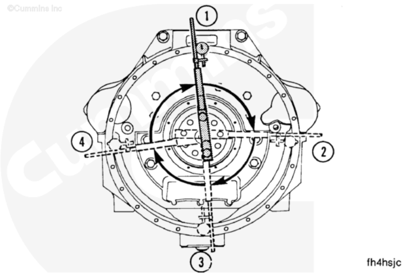
Attach an indicator as shown. Position the indicator at the 12-o'clock position. Adjust the dial until the needle points to zero.



Flywheel Housing - Face Runout				
	mm		in	
Number 0		0.25	MAX	0.010
Number 00		0.30	MAX	0.012

Record the indicator reading at three different points; for example, 3 o'clock, 6 o'clock, and 9 o'clock.

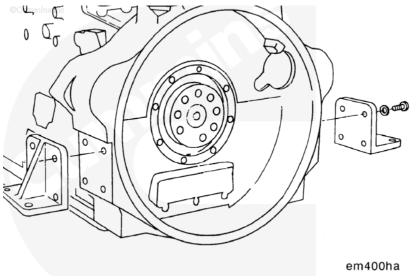
Turn backward to the original position. Make sure the needle still points to zero. Determine the total indicator runout.



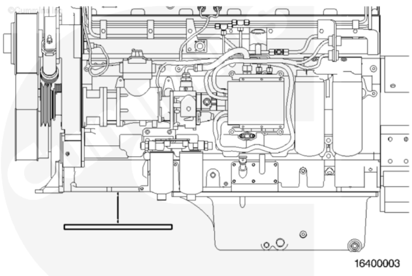
Example of Total Indicator Run Out	
Position	**Indicator Reading**
3 o'clock	0.00 mm [0.00 in]
6 o'clock	-0.08 mm [-0.003 in]
9 o'clock	0.05 mm [0.002 in]
Total Indicator Run Out = 0.13 mm [0.005 in]	

Install the rear engine mounts to the flywheel housing.

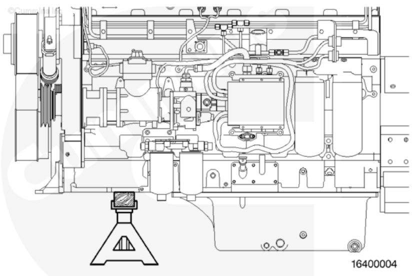
Torque Value: 550 n•m [406 ft-lb]



Install the adapter cover plate or oil sump, whichever is in the front position.



Lift the engine. Remove the jack stands and supports, and lower the engine.



Finishing Steps

Industrial and Power Generation

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first, and attach the negative (-) battery cable last.

- Install the rear crankshaft seal. Refer to Procedure 001-024 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-024-tr.html)
- Fill the engine with lubricating oil. Refer to Procedure 007-037 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-037-tr.html)
- Install the flywheel. Refer to Procedure 016-005 in Section 16. (/qs3/pubsys2/xml/en/procedures/56/56-016-005-tr.html)
- Install the starting motor. Refer to Procedure 013-020 in Section 13. (/qs3/pubsys2/xml/en/procedures/56/56-013-020-tr.html)
- Install the engine speed sensor. Use the following procedure in the QSK19, QSK23, QSK45, QSK60, and QSK78 Electronic Control System Troubleshooting and Repair Manual, Bulletin 3666113. Refer to Procedure 019-042 in Section 19. (/qs3/pubsys2/xml/en/procedures/19/19-019-042.html)
- Install the transmission, clutch, and all related components. See equipment manufacturer service information.
- Connect the batteries. See equipment manufacturer service information.
- Connect the air starter, if equipped. See equipment manufacturer service information.

with Electronically Actuated Injector

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- Fill the engine with lubricating oil. Refer to Procedure 007-037 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-037-tr.html)
- Install the flywheel. Refer to Procedure 016-005 in Section 16. (/qs3/pubsys2/xml/en/procedures/56/56-016-005-tr.html)
- Install the starting motor. Refer to Procedure 013-020 in Section 13. (/qs3/pubsys2/xml/en/procedures/56/56-013-020-tr.html)
- Install the engine speed sensor. Use the following procedure in the QSK38, QSK50 and QSK60 (CM850 Modular Common Rail System) Troubleshooting and Repair Manual, Bulletin 4021533. Refer to Procedure 019-042 in Section 19. (/qs3/pubsys2/xml/en/procedures/05/05-019-042.html)
- Install the rear bank wiring harness. Use the following procedure in the QSK38, QSK50 and QSK60 (CM850 Modular Common Rail System) Troubleshooting and Repair Manual, Bulletin 4021533. Refer to Procedure 019-043 in Section 19. (/qs3/pubsys2/xml/en/procedures/05/05-019-043.html)
- Install the turbocharger oil drain lines. Refer to Procedure 010-034 in Section 10. (/qs3/pubsys2/xml/en/procedures/56/56-010-034-tr.html) Refer to Procedure 010-035 in Section 10. (/qs3/pubsys2/xml/en/procedures/56/56-010-035-tr.html)
- Install the transmission, clutch, and all related components. See equipment manufacturer service information.
- Connect the batteries. See equipment manufacturer service information.
- Connect the air starter, if equipped. See equipment manufacturer service information.

QSK60 Marine Applications

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- Install the rear crankshaft seal. Refer to Procedure 001-024 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-024-tr.html)

- Fill the engine with lubricating oil. Refer to Procedure 007-037 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-037-tr.html)
- Install the flywheel. Refer to Procedure 016-005 in Section 16. (/qs3/pubsys2/xml/en/procedures/56/56-016-005-tr.html)
- Install the starting motor. Refer to Procedure 013-020 in Section 13. (/qs3/pubsys2/xml/en/procedures/56/56-013-020-tr.html)
- Install the transmission, clutch, and all related components. See equipment manufacturer service information.
- Install the turbocharger air inlet connections. Refer to Procedure 010-130 in Section 10. (/qs3/pubsys2/xml/en/procedures/56/56-010-130-tr.html)
- Install the air cleaner assembly. Refer to Procedure 010-013 in Section 10. (/qs3/pubsys2/xml/en/procedures/56/56-010-013-tr.html)
- Connect the batteries. See equipment manufacturer service information.
- Connect the air starter, if equipped. See equipment manufacturer service information.

Last Modified: 16-Jun-2016